

Analyzing the Connection Between Early-Childhood Participation in WIC and Later Academic Performance

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Introduction

The Special Supplemental Nutrition Program for Women, Infants, and Children (WIC) is one of the largest programs for nutrition assistance in the United States. The program is designed to support low-income pregnant women, infants, and young children by improving access to nutritious foods, nutrition education, and healthcare referrals. Even though the program has been widely studied, much of the existing literature focuses on short-term outcomes, such as immediate health and cognitive benefits during early childhood. There is less knowledge about whether these early advantages translate into long-term educational development, particularly for children who participate in WIC between the ages of one and five. Understanding these longer-run effects is crucial for evaluating the efficiency and societal value of the program. Prior research provides strong evidence that WIC participation improves early-life outcomes. Changes to WIC food packages have been related with lower childhood obesity rates, suggesting that improved nutritional quality has meaningful effects on children's health trajectories (Daepp et al., 2019). WIC participation has also been associated with behavioral problems, earlier prenatal and early childhood exposure reduces behavioral problems, highlighting the program's role in supporting socio-emotional development (Chorniy, Currie, & Sonchak, 2020). In combination, these findings demonstrate that WIC has significant short-run benefits; however, they leave open the question of whether these early gains persist as children progress through the educational system.

This paper would explore how to address the gap by examining the long-term effects of WIC participation at young ages on educational development. Our research question asks, what is the effect of participating in WIC during early childhood on long-term educational outcomes? This question seeks to directly broaden public health and socioeconomic concerns, including child poverty, education inequality, and integration mobility. Children from low-income households face persistent disadvantages that can compound over time, making early interventions especially important. If WIC participation generates lasting educational benefits, the program may serve as a cost-effective tool for reducing inequality and improving long-run economic outcomes.

Despite WIC's benefits, WIC enrollment rates remain lower than eligibility rates, raising concerns about access and program design. Barriers to participation include food package inflexibility, limited funding for fruits and vegetables, stigma associated with using Electronic Benefit Transfer (EBT) cards, and difficulties locating WIC-approved products. In addition, lack of awareness among eligible mothers further suppresses participation. A review by Davis, Leavitt, and Chaucer (2022) highlights that these structural and informational barriers reduce enrollment and continued participation, even among families who could benefit most from the program. This paper thus aims to contribute to literature about the benefits of WIC by focusing on the potential long term academic effects of the program on children ages 1-4 in efforts to signify whether there is a greater perceived marginal benefit to WIC participation for youth not previously considered by present biased mothers.

Literature Review

The Benefits of WIC:

WIC has proven to serve as a cost-effective investment that improves nutrition for low-income families by reducing the risk of adverse birth outcomes. Studies support that prenatal WIC participation lowers the risk of infant mortality, develops healthier diets and feeding practices, and overall is correlated to lower obesity rates for child participants (Carlson and Neuberger). Though evidence provides numerous positive results from WIC participation, there remains a large gap between eligibility rates and those enrolled. As of 2021, less than 52% of eligible individuals participated in WIC, with over 4 million participants from children aged 1-4 missing out on WIC health and developmental benefits (Neuberger et al.). One of the potential explanations for lack of participation discusses costs attributed to WIC. Many findings attribute perceived compliance costs among the main reasons mothers see less value in maintaining WIC, of which costs are perceived too high typically after formula stops being supplied or socioeconomic status changes (Lora et al). Compliance costs consider the troubles of making time for and managing appointments, bringing in children and signing documents (Barnes et al.). Strategies to increase participation in the program are therefore not related to funding, but rather outreach where misinformation or a lack of information is implied.

WIC affect on Academic Achievements

Though not many significant studies have been conducted towards the long-term effects of WIC, small bodies of research do provide evidence that WIC participation can be linked to cognitive development and academic achievement. For example, one study by Margot Jackson analyzed information from the ECLS-B, which provided data on WIC participation and used the

Child Development Study [CDS] to conduct a longer examination between WIC and academic achievement. Jackson first estimated a relationship between WIC participation and cognitive development at 2 years old. Results demonstrated prenatally exposed 2 year olds scored higher [0.062] standard deviations on the Bayley Mental Development [BMD] assessment than their peers and siblings who did not receive WIC in utero. Next the study utilized the CDS of the panel study of income dynamics PSID to conduct a long term analysis on broad reading and math scores. Though limited to a smaller sample, scores from the Woodcock Johnson Revised Reading and Math tests also suggested children who receive prenatal/early childhood exposure to WIC perform significantly better on reading assessments, up to 0.3 of the standard deviation, than their siblings who do not participate (Jackson). The relationship found between a prenatal sibling participating in WIC having overall better academic results than their sibling [who did not participate] corresponds to other studies which have found participant siblings have a 5% lower probability of being diagnosed with ADHD and a 5.1% lower probability of being diagnosed with more mental health conditions later on than their non participating siblings (Currie). Using birth, medicaid, and school records in South Carolina, the study also proposed a potential relationship between having less mental conditions in later school years with grade repetition, finding that WIC participation could potentially lower frequency of grade repetition by almost 8% (Currie). Lastly, we consider research regarding the relationship between food insecurity on long term academic outcome, since there have been proven associations between WIC participation and food security in the short term (Cook and Frank). Several studies use data on responses to USDA food sufficiency questions and compare it between children's health, school performance, and psychological functioning - providing a series of results. One study found food insufficiency associated with higher mineral deficiencies which increased the

likelihood of experiencing sickness for children 1-5 years. Another demonstrated children 6-11 and teens facing food insufficiencies had lower arithmetic scores and were more likely to repeat a grade, see a psychologist, experience suspension from school, or have a harder time socializing. And numerous more studies conclude statistically significant increases in depression and impaired functioning amongst children in food insufficient households or experiencing higher risks of hunger (Cook and Frank).

Constraints & Bias

However, primarily studies which attempt to directly measure the effects of WIC, faced significant limitations - the largest including their usage of the sibling data comparisons and the lack of available data for WIC due to privacy or geographical constraints. Researchers utilizing the mother fixed effect models restrict samples to children with siblings. This therefore limits the amount of usable data and comparisons since they were unable to address single children or other potential reasoning for sibling differences which could have led to an over or underestimation of WIC benefits. Lack of available data also stems from the amount of information required to find when and if children participated in WIC, which is why many studies are state or time period specific. However, studies that incorporate data from prior to 2009 in their measurements are not accurately measuring the potential benefits from WIC which had a large food package change then. Therefore, though large amounts of literature are able to demonstrate positive effects of WIC on birth outcomes and potential future health, research seems to lack strong evidence towards long term benefits that spillover from providing children the potential for healthier lifestyles. The intention of our study is thus to provide a stronger alternative, and easier to replicate, manner of analyzing the effects of WIC on academic achievement which avoids previous studies data constraints. With this provided, findings

demonstrating a significant relationship between WIC coverage rates and educational outcomes could provide insight on a potential positive externality not considered by eligible WIC participants who leave early to misinformation or increased socioeconomic status, which overall provides stronger evidence to combat compliance costs by adding long term benefits.

Proposed Methodology and Data Collection:

The purpose of our research is to investigate whether improved state-level WIC Coverage Rate for eligible children age 4 from 2011 to 2013 is associated with stronger fourth-grade educational outcomes from 2017 to 2019. Our proposed methodology is to construct a two-period panel regression that includes two outcome years, two cohorts, and data from each of the fifty states. This econometric strategy takes inspiration from Hoynes et. al's research entitled "Long-Run Impacts of Childhood Access to the Safety Net," which analyzes how early-childhood cohort exposure to the Food Stamp Program rollout between 1961 and 1975 impacts wages and educational attainment (Hoynes et. al 2016). Although our research does not utilize differences-in-differences causal identification, we also intend to incorporate variation in cohort exposure and regions to analyze later educational outcomes. Below is the proposed economic specification:

$$Score_{s,t} = \beta CoverageRate_{s,t-6} + \gamma X_{s,t-6} + \alpha_s + \lambda_t + \epsilon_{s,t}$$

s=state

t=2017, 2019

t-6=2011, 2013

(WIC Coverage
years)

CoverageRate is the key independent variable that measures the state-level WIC Coverage Rate among four-year-olds (calculated by dividing the number of children age 4 participating in WIC by the total number of four-year old eligible children). The first assigned cohort for coverage rate exposure is in 2011, and the second assigned cohort in 2013. The Score outcome variables are measured six years after the CoverageRate for both cohorts, when they are estimated to be age 10 and eligible to complete the country-wide fourth-grade National Assessment of Educational Progress (NAEP assessment). The dependent variable, Score, measures state-level NAEP fourth-grade average reading or math scores in 2017 and 2019, with reading and math scores tested separately. The coefficient of interest, β , measures whether within-state changes in the WIC Coverage Rate from the 2011 cohort to the 2013 cohort are associated with within-state changes in 4th-grade NAEP scores. Endogeneity concerns arise because unobserved factors may be causing biased estimates of the relationship between WIC coverage rates and educational outcomes. In order to address this, state fixed effects, year fixed effects, and time-varying state-level controls will be included in the regression. α , state fixed effects, addresses endogeneity concerns by controlling for differences in stable state characteristics that also influence educational outcomes, such as institutions, economic conditions, and long-term demographics. λ , year fixed effects, controls for national shocks correlated with education, such as federal policy changes, economic downturns, and educational reforms. Time-varying factors that likely cause differences in WIC Coverage rates will also be included, such as changes in poverty, household income, urbanization, and educational funding.

WIC Coverage Rate data for both exposure cohorts will be obtained from the 2011 and 2013 *National and State-level Estimates of Special Supplemental Nutrition Program for Women, Infants, and Children* reports published by the USDA. The reports contain state-level estimates

of WIC Coverage Rates that include ages 1-4, infants, and Coverage Rates specifically for eligible children age 4. They obtained yearly estimates utilizing “a methodology developed by the Urban Institute that apportions the national figures using data from the American Community Survey” (U.S. Department of Agriculture, Food and Nutrition Service). Since eligibility may change throughout the year, the USDA used the longitudinal Survey of Income and Program Participation (SIPP) to predict the share of months a child was eligible and aggregated the probabilities to obtain state-level estimates. While the reports are widely used for research purposes and derived from high-quality, reliable methods, minor deviations from precise eligibility rates are possible. State-level average fourth-grade NAEP reading and math scores are published biennially and will be obtained from *The Nation’s Report Card: State Profiles* website by the National Center for Education Statistics. NAEP writing and science assessments are also conducted, but data is missing for many states, so they were not included as outcome variables. The 2011-2017 and 2013-2019 cohorts were selected to avoid endogeneity concerns regarding COVID-19 educational disruptions. Additionally, the USDA modified their methodology for estimating WIC Coverage Rates beginning in 2018, so later reports are not comparable to earlier reports. Data for state-level exposure year controls, including median household income, child poverty rate, unemployment rate, racial composition, and the percent of urban and rural populations are available from the U.S. Census Bureau’s *American Community Survey*. Finally, data for the outcome-year per-pupil K-12 spending control will be obtained from the U.S. Census Bureau’s *Public-Elementary-Secondary Education Finance Data*.

Heterogeneity- School Funding:

We will also be interested in exploring how the relationship between WIC Coverage and educational outcomes interacts with heterogeneity in school funding, as shown in the regression below:

$$Score_{s,t} = \beta CoverageRate_{s,t-6} + \beta_2(CoverageRate_{s,t-6} * LowSpending_s) + \gamma X_{s,t-6} + \alpha_s + \lambda_t + \epsilon_{s,t}$$

This additional analysis will maintain the key components of the baseline regression. However, it will assign a LowSpending binary indicator that if the state's average per-pupil K-12 spending falls below the national median across 2017 and 2019, and interact the LowSpending binary indicator with CoverageRate. β_2 , the new coefficient of interest, measures the differential impact of the relationship between WIC Coverage and NAEP scores on lower-spending states. If β_2 is positive and statistically significant, this indicates a stronger association in low-spending states, and a possible role of nutrition in substituting for or overcoming fewer school resources.

However, a significant negative β_2 coefficient indicates a weaker association in low-spending states, and that WIC potentially benefits educational outcomes more in environments that already have strong resources.

Limitations and Potential Important Insights:

We will not be able to establish a causal relationship because we lack sources of exogenous variation in WIC participation during the time-frame of the study, and privacy restrictions prevent us from accessing individual-level data. Another acknowledged limitation is that linking WIC cohort exposure to state-level educational outcomes is an estimation rather than exact, and cannot perfectly account for children repeating or skipping grade-levels. Additionally, we are only able to compare two cohorts due to methodological constraints and endogeneity

concerns discussed in the *Proposed Methodology* section. However, we will estimate meaningful, robust within-state associations between changes in WIC Coverage Rate and average NAEP score outcomes by controlling for state and year fixed effects, as well as additional time-varying controls. Our research will offer several important insights for policymakers and families eligible for WIC participation. A significant positive relationship between improvements in WIC Coverage Rates and average NAEP scores will further demonstrate the importance of early nutritional environments on later educational outcomes, and begin exploration into an understudied additional benefit of WIC participation. Since Math and Reading NAEP score outcomes are tested separately, the results will demonstrate whether WIC may impact quantitative or qualitative skills differently. Furthermore, examining the interactions between nutritional programs, school resources, and educational outcomes is important for developing effective policies, particularly when addressing disparities in lower-resource educational environments. We intend for this study to encourage outreach efforts to improve WIC uptake, as well as act as an impetus for further research that utilizes restricted-access administrative data to more precisely estimate WIC's diverse long-term impacts.

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